ECEN-601: Homework 2

Due on September 19, 2025 at 5:00 pm

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Problem 1

Negate the following statement: For every real number $\varepsilon > 0$ there exists a positive integer k such that all positive integers n, it is the case that $|a_n - k^2| < \varepsilon$.

Solution

Problem 2

Let $f : X \to Y$, $A, B \subseteq X$, and $C, D \subseteq Y$. Show that f^{-1} preserves inclusions, unions, and intersections.

[Hint: Recall that $f(A) \triangleq \{f(x) \mid x \in A\}$ and $f^{-1}(C) \triangleq \{x \in X \mid f(x) \in C\}$.]

(a) Show that $C \subseteq D \implies f^{-1}(C) \subseteq f^{-1}(D)$.

(b) Show that $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$.

(c) Show that $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$.

Show that f preserves inclusions and unions, but not intersections.

(a) Show that $A \subseteq B \implies f(A) \subseteq f(B)$.

(b) Show that $f(A \cup B) = f(A) \cup f(B)$.

(c) Show that $f(A \cap B) \subseteq f(A) \cap f(B)$. Give an example where equality does not occur.

Solution

Problem 3

Let $\underline{x} = (x_1, \ldots, x_n)$, $\underline{y} = (y_1, \ldots, y_n) \in \mathbb{R}^n$ and consider the function ρ given by

$$\rho(\underline{x}, \underline{y}) = \max |x_1 - y_1|, \cdots, |x_n - y_n|.$$

Show that ρ is a metric.

Solution

Problem 4

Problem 5

Problem 6

Problem 7